PROJECT ATO / INTEGRATED MODULAR CIRCUIT SYSTEM

Eight functional boards; actual mapped electrical connections and physical tap headers.

Reference namespaces preserve frozen source identities. Read integrated-pcb-wiring.csv for the harness.

Common driver rearm and independent supervisor phase rearm are deliberately separated.

Digital artificial-load concept. Geometry and ERC do not establish implantation or measured performance.

monitoring-cell

File: monitoring-cell/monitoring_cell.kicad_sch

current-driver

File: current-driver/current_driver.kicad_sch

recording-front-end

File: recording-front-end/recording_front_end.kicad_sch

contact-router

File: contact-router/contact_router.kicad_sch

supervision

File: supervision/supervision.kicad_sch

power

File: power/power.kicad_sch

monitoring-transport

File: monitoring-transport/monitoring_transport.kicad_sch

controller-carrier

File: controller-carrier/controller_carrier.kicad_sch

Project ATO

Sheet: /

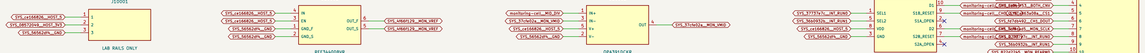
File: ATO_integrated.kicad_sch

Title: ATO / Integrated modular SCCI electrical system

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/80

ARTIFICIAL ELECTRICAL LOADS ONLY / FIRST MONITORING SLICE

No current driver. HV router. differential compliance circuit. Independent watchdog. MCU or PCB is completed here.











V_REF=4.096V; V_MID=2.048V. Low INT_RUN resets: high opens reset for continuous signed integration.

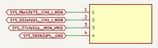
I_LO=1.04757V / I_HI=3.04843V nominal (−+/−25.01mA). Threshold is NOT a proven complete-waveform cap.

EXT_SUP_OK must stay LOW until external rails/reference/watchdog/compliance/duration checks pass.

Two INA241 channels alone: up to32mW atSV. Input bias25.45uA/input is an unresolved leakage/bypass term.

Named global nets connect sheets; connectors remain explicit external design contracts. Read README before interpretation.





Source0_monitor

File: channel0.kicad_sch

Source1_monitor

File: channel1.kicad_sch

Integrated_component_taps

File: integration_taps.kicad_sch

ARTIFICIAL LOADS ONLY; native schematic slice, not a completed implant

Project ATO

Sheet: /monitoring-cell/

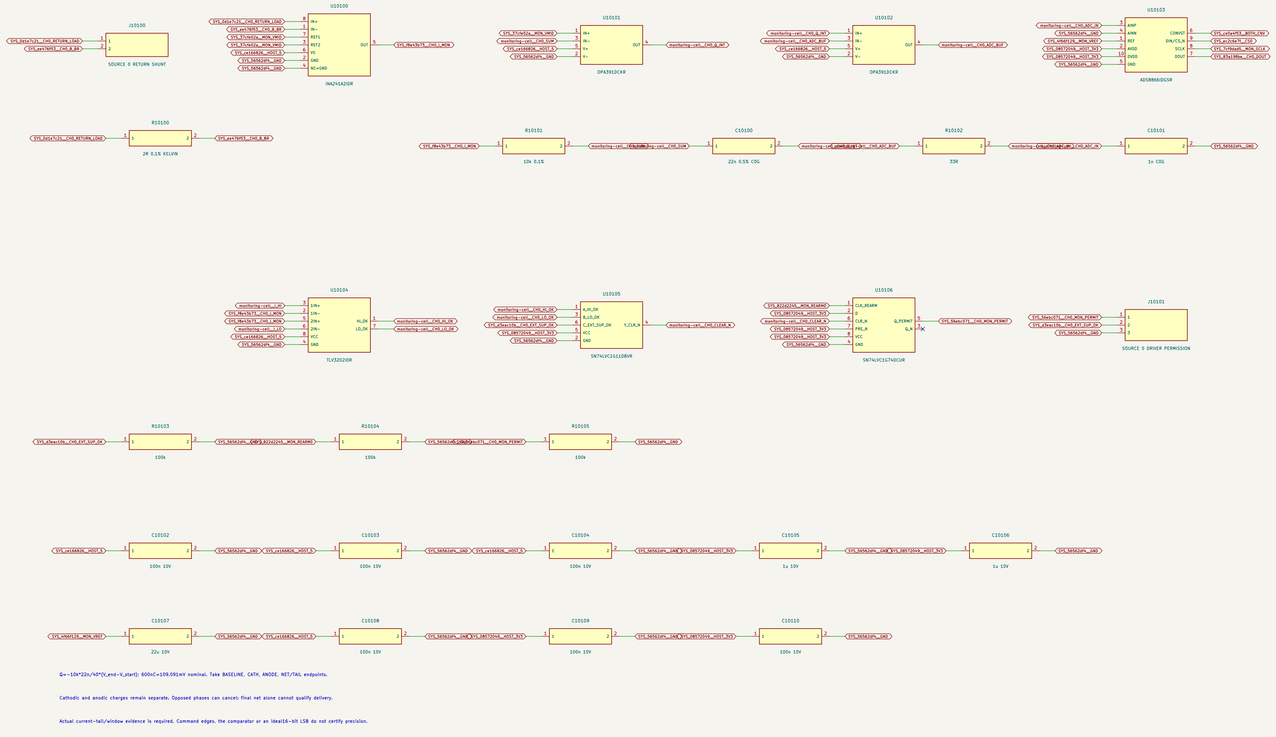
File: monitoring_cell.kicad_sch

Title: ATO S4 / Shared reference and interfaces

Size: A1 Date: 2026-10-12	**Rev: 0.2 DCK**
KiCad E.D.A. 10.0.6	Id: 2/80

ARTIFICIAL ELECTRICAL LOADS ONLY / FIRST MONITORING SLICE

No current driver. HV router. differential compliance circuit. Independent watchdog. MCU or PCB is completed here.



Qₙ=10k*22n/40*(V_end−V_start): 600nC=109.091mV nominal. Take BASELINE, CATH. ANODE, NET/TAIL endpoints.

Cathodic and anodic charges remain separate. Opposed phases can cancel: final net alone cannot qualify delivery.

Actual current-tail/window evidence is required. Command edges. the comparator or an ideal16-bit LSB do not certify precision.

CLRN = HI_OK AND LO_OK AND EXT_SUP_OK. Any low latches Q off: rising REARM admits only while CLRN is high.

ARTIFICIAL LOADS ONLY; native schematic slice, not a completed implant

Project ATO

Sheet: /monitoring-cell/Source0_monitor/

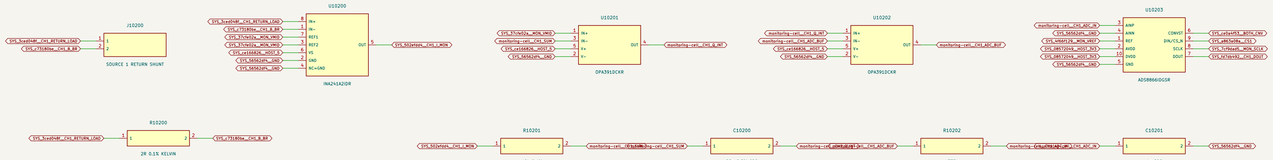
File: channel0.kicad_sch

Title: ATO S4 / Source 0: current and phase charge

Size: A1	Date: 2026-10-12	**Rev: 0.2 DCK**
KiCad E.D.A. 10.0.6		Id: 3/80

ARTIFICIAL ELECTRICAL LOADS ONLY / FIRST MONITORING SLICE

No current driver. HV router. differential compliance circuit. independent watchdog. MCU or PCB is completed here.



Q=-10k*22n/40*(V_end-V_start): 600nC=109.091mV nominal. Take BASELINE, CATH. ANODE, NET/TAIL endpoints.

Cathodic and anodic charges remain separate. Opposed phases can cancel final net alone cannot qualify delivery.

Actual current-tail/window evidence is required. Command edges. the comparator or an ideal16-bit LSB do not certify precision.

CLRN = HI_OK AND LO_OK AND EXT_SUP_OK. Any low latches Q off: rising REARM admits only while CLRN is high.

ARTIFICIAL LOADS ONLY; native schematic slice, not a completed implant

Project ATO

Sheet: /monitoring-cell/Source1_monitor/

File: channel1.kicad_sch

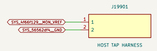
Title: ATO S4 / Source 1: current and phase charge

Size: A1	Date: 2026-10-12	**Rev: 0.2 DCK**
KiCad E.D.A. 10.0.6		Id: 4/80

INTEGRATED COMPONENT-TAP HARNESS / monitoring-cell

Physical headers expose actual earlier component taps. Original connectors remain available.

Only named electrical nets connect. AFE, HOST and companion grounds remain distinct.

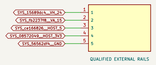


Artificial-load digital design. Added wiring capacitance and connector integrity require measurement.

Project ATO
Sheet: /monitoring-cell/integrated_component_taps/
File: integration_taps.kicad_sch
Title: ATO / monitoring-cell integration taps
Size: A1 | Date: 2026-10-14 | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 5/80

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.







Source0_analog

File: source0.kicad_sch

Source0_permission

File: permission0.kicad_sch

Source1_analog

File: source1.kicad_sch

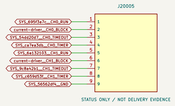
Source1_permission

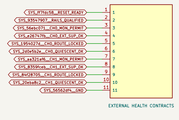
File: permission1.kicad_sch













DAC initialization before READY: GAIN[5:0]=0x0103, reference divider 2 and both output buffer gains 2 => nominal 2.5 V FSR.

Code 7864 -> 2.999878 mA nominal through 100 ohm. Zero code is not terminal disconnection. Read README and external contracts.

RESET_READY held LOW through startup; REARM is an explicit pulse with REQUEST/TIMER LOW and independently observed quiescence.





Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /current-driver/

File: current_driver.kicad_sch

Title: ATO S4 / Dual setpoint and external contracts

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/80

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

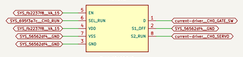
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

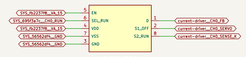


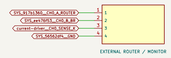




Current path: HV -> upper switch -> artificial pair load + external 2 ohm monitor -> lower switch -> BSP89 -> 100 ohm -> GND.







RUN low grounds gate and closes local unity feedback. RUN high selects current loop. DNP compensation needs a transient study.

























Bridge uses captured direction: all four commanded off during inhibit. Analog rail clamps, input bias and stuck switches remain real paths.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /current-driver/Source0_analog/

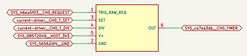
File: source0.kicad_sch

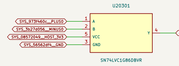
Title: ATO S4 / Source 0: regulated sink and polarity bridge

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 7/80

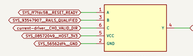
ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

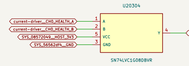
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

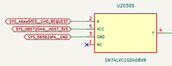








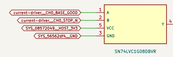


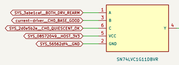


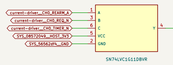




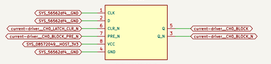


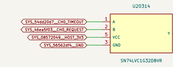


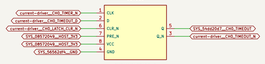


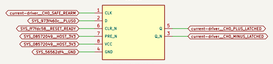


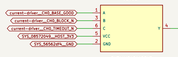




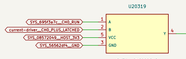


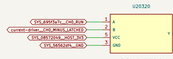












































































BASE_GOOD=MON & EXT & READY & RAILS & XOR(raw directions). Raw REQUEST clocks TIMER; SAFE_REARM preloads direction while off.

Normal request end consumes BLOCK; TIMEOUT records held request at timer fall. REARM requires request/timer LOW + QUIESCENT_OK.

Static/event checks do not qualify asynchronous hazards, supply ramps, propagation delay or real current tails. No automatic retry.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /current-driver/Source0_permission/

File: permission0.kicad_sch

Title: ATO S4 / Source 0: independent duration and sticky admission

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 8/80

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

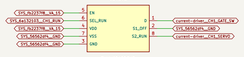
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

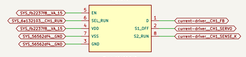


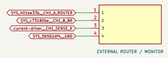




Current path: VH -> upper switch -> artificial pair load + external 2 ohm monitor -> lower switch -> BSP89 -> 100 ohm -> GND.







RUN low grounds gate and closes local unity feedback; RUN high selects current loop. DNP compensation needs a transient study.

























Bridge uses captured direction: all four commanded off during inhibit. Analog rail clamps, input bias and stuck switches remain real paths.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /current-driver/Source1_analog/

File: source1.kicad_sch

Title: ATO S4 / Source 1: regulated sink and polarity bridge

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 9/80

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.









BASE_GOOD=MON & EXT & ROUTE & READY & RAILS & XOR(raw directions). Raw REQUEST clocks TIMER; SAFE_REARM preloads direction while off.

Normal request end consumes BLOCK; TIMEOUT records held request at timer fall. REARM requires request/timer LOW + QUIESCENT_OK.

Static/event checks do not qualify asynchronous hazards, supply ramps, propagation delay or real current tails. No automatic retry.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /current-driver/Source1_permission/

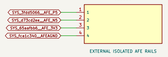
File: permission1.kicad_sch

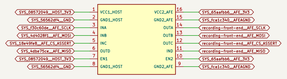
Title: ATO S4 / Source 1: independent duration and sticky admission

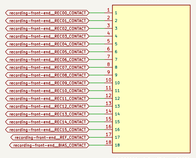
Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 10/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

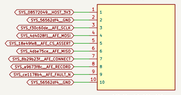
Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

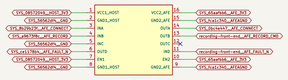




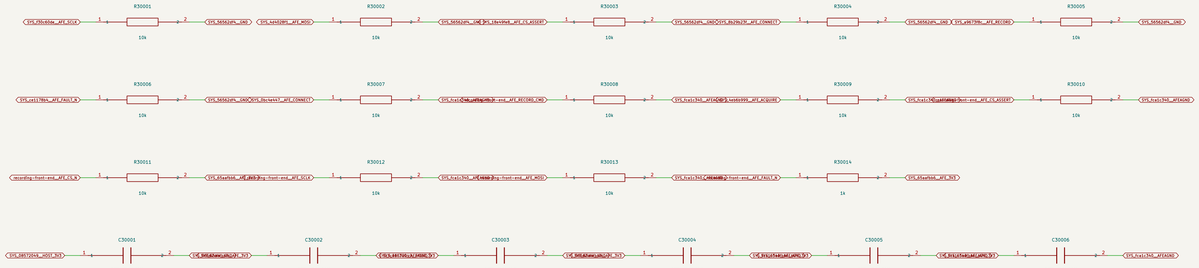












HOSTGND and AFEAGND are separate domains. Both receive valid regulated supplies externally; the digital barrier does not generate power.

acquisition



File: acquisition.kicad_sch

bank0



File: bank0.kicad_sch

bank1



File: bank1.kicad_sch

bank2



File: bank2.kicad_sch

bank3



File: bank3.kicad_sch

reference

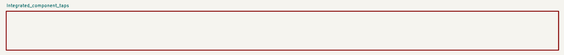


File: reference.kicad_sch

CS_ASSERT low -> Island CS_N high. CONNECT low disconnects HV paths: ACQUIRE = CONNECT & RECORD, so O1 remains clamped.

Driver phase admission needs independently observed ISOLATED_READY; neither commands nor fault flags establish voltage or delivery evidence.

integrated_component_taps



File: integration_taps.kicad_sch

Native digital circuit slice: artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /recording-front-end/

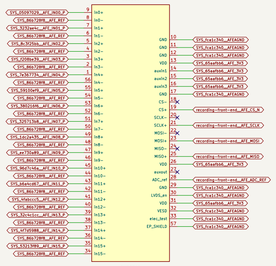
File: recording_front_end.kicad_sch

Title: ATO S4 / isolated SPI, control and supplied power

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 11/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.



Every minus input shares AFE_REF. This is 16 differential streams from 18 functional recording ports, not 32 independent recording electrodes.









CMOS mode: LVDS_en=0: negative SPI pins NC; unused aux inputs at VDD: avaout NC; elec_test and VESD at AFEAGND.

Configure, calibrate and verify readback via SPI before valid data. 4MHz / 5us command slots are the revised isolated-link study baseline.

ADC response is pipelined: physical sample timestamps and invalid blanking/recovery intervals must be retained in software.

Native digital circuit slice: artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /recording-front-end/acquisition/

File: acquisition.kicad_sch

Title: ATO S4 / RHD2216 / sixteen sampled differential streams

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 12/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

R30100
recording-front-end_REC00_CONTACT
1
2
recording-front-end_REC00_SERIES
4.99k 0.1%
R30101
recording-front-end_REC01_CONTACT
1
2
recording-front-end_REC01_SERIES
4.99k 0.1%
R30102
recording-front-end_REC02_CONTACT
1
2
recording-front-end_REC02_SERIES
4.99k 0.1%
R30103
recording-front-end_REC03_CONTACT
1
2
recording-front-end_REC03_SERIES
4.99k 0.1%
R30200
recording-front-end_REC00_PROT
1
2
SYS_fca12e2_AFEAGND
10M
R30201
recording-front-end_REC01_PROT
1
2
SYS_fca12e2_AFEAGND
10M
R30202
recording-front-end_REC02_PROT
1
2
SYS_fca12e2_AFEAGND
10M
R30203
recording-front-end_REC03_PROT
1
2
SYS_fca12e2_AFEAGND
10M
U30100
SYS_65ae9bb6_AFE_IN3
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da0506a_AFE_IP3
SYS_d7b1d3ee_AFE_AB3
SYS_fca12e2_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
S_RHD
S1_CLAMP
S2_PROTECTED
SYS_35597529_AFE_IN03_P
SYS_fca12e2_AFEAGND
recording-front-end_REC00_PROT
TMUX8219DGKR
U30101
SYS_65ae9bb6_AFE_IN3
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da0506a_AFE_IP3
SYS_d7b1d3ee_AFE_AB3
SYS_fca12e2_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
S_RHD
S1_CLAMP
S2_PROTECTED
SYS_3233aeba_AFE_IN03_P
SYS_fca12e2_AFEAGND
recording-front-end_REC01_PROT
TMUX8219DGKR
U30102
SYS_65ae9bb6_AFE_IN3
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da0506a_AFE_IP3
SYS_d7b1d3ee_AFE_AB3
SYS_fca12e2_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
S_RHD
S1_CLAMP
S2_PROTECTED
SYS_8c3f2f0b_AFE_IN02_P
SYS_fca12e2_AFEAGND
recording-front-end_REC02_PROT
TMUX8219DGKR
U30103
SYS_65ae9bb6_AFE_IN3
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da0506a_AFE_IP3
SYS_d7b1d3ee_AFE_AB3
SYS_fca12e2_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
S_RHD
S1_CLAMP
S2_PROTECTED
SYS_f2088e39_AFE_IN03_P
SYS_fca12e2_AFEAGND
recording-front-end_REC03_PROT
TMUX8219DGKR
C30100
SYS_35597529_AFE_IN03_P
1
2
SYS_fca12e2_AFEAGND
470p C0G 50V
C30101
SYS_3233aeba_AFE_IN03_P
1
2
SYS_fca12e2_AFEAGND
470p C0G 50V
C30102
SYS_8c3f2f0b_AFE_IN02_P
1
2
SYS_fca12e2_AFEAGND
470p C0G 50V
C30103
SYS_f2088e39_AFE_IN03_P
1
2
SYS_fca12e2_AFEAGND
470p C0G 50V
C30200
SYS_9da0506a_AFE_IP3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30201
SYS_9da0506a_AFE_IP3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30202
SYS_9da0506a_AFE_IP3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30203
SYS_9da0506a_AFE_IP3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30300
SYS_d7b1d3ee_AFE_AB3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30301
SYS_d7b1d3ee_AFE_AB3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30302
SYS_d7b1d3ee_AFE_AB3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
C30303
SYS_d7b1d3ee_AFE_AB3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V

Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2; S1 grounds RHD until ACQUIRE is admitted.

C30020
SYS_9da0506a_AFE_IP3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V
U30020
SYS_35e1ee47_AFE_CONNECT
SYS_35e1ee47_AFE_CONNECT
SYS_35e1ee47_AFE_CONNECT
SYS_35e1ee47_AFE_CONNECT
SYS_9da0506a_AFE_IP3
SYS_d7b1d3ee_AFE_AB3
SYS_fca12e2_AFEAGND
EP_SHIELD
SEL1
SEL2
SEL3
SEL4
VDD
VSS_NEG
GND
S1_CONTACT
S1_INTERNAL
S2_CONTACT
S2_INTERNAL
S3_CONTACT
S3_INTERNAL
S4_CONTACT
S4_INTERNAL
FF_N
recording-front-end_REC00_SERIES
recording-front-end_REC00_PROT
recording-front-end_REC01_SERIES
recording-front-end_REC01_PROT
recording-front-end_REC02_SERIES
recording-front-end_REC02_PROT
recording-front-end_REC03_SERIES
recording-front-end_REC03_PROT
recording-front-end_AFE_FAULT_N
TMUX7612FRRPR
C30030
SYS_d7b1d3ee_AFE_AB3
1
2
SYS_fca12e2_AFEAGND
100n X7R 16V

470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared-reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain
Project ATO
Sheet: /recording-front-end/bank0/
File: bank0.kicad_sch
Title: ATO S4 / Inputs 00–03 / protected recording
Size: A1 | Date: 2026-10-13 | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 13/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

R30104
4.99k 0.1%
R30105
4.99k 0.1%
R30106
4.99k 0.1%
R30107
4.99k 0.1%
R30204
10M
R30205
10M
R30206
10M
R30207
10M
U30104
TMUX8219DGKR
U30105
TMUX8219DGKR
U30106
TMUX8219DGKR
U30107
TMUX8219DGKR
C30104
470p C0G 50V
C30105
470p C0G 50V
C30106
470p C0G 50V
C30107
470p C0G 50V
C30204
100n X7R 16V
C30205
100n X7R 16V
C30206
100n X7R 16V
C30207
100n X7R 16V
C30304
100n X7R 16V
C30305
100n X7R 16V
C30306
100n X7R 16V
C30307
100n X7R 16V

Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2; S1 grounds RHD until ACQUIRE is admitted.

C30021
100n X7R 16V
U30021
TMUX7412FRBPR
C30031
100n X7R 16V

470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared-reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /recording-front-end/bank1/

File: bank1.kicad_sch

Title: ATO S4 / Inputs 04–07 / protected recording

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 14/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

R30108

recording-front-end_RECOB_CONTACT

4.99k 0.1%

recording-front-end_RECOB_SERVE1

R30109

recording-front-end_RECOA_CONTACT

4.99k 0.1%

recording-front-end_RECOA_SERVE1

R30110

recording-front-end_RECS3_CONTACT

4.99k 0.1%

recording-front-end_RECS3_SERVE1

R30111

recording-front-end_RECS2_CONTACT

4.99k 0.1%

recording-front-end_RECS2_SERVE1

R30208

recording-front-end_RECOB_PROT

10M

SYS_Vss1v2kS_AFEAGND

R30209

recording-front-end_RECOA_PROT

10M

SYS_Vss1v2kS_AFEAGND

R30210

recording-front-end_RECS3_PROT

10M

SYS_Vss1v2kS_AFEAGND

R30211

recording-front-end_RECS2_PROT

10M

SYS_Vss1v2kS_AFEAGND

U30108

SYS_EGeeYbbK_AFE_RX3

SYS_hmS69999_AFE_ACQUIRE

SYS_9tpdO5bk_AFE_PS

SYS_d7bCd2ew_AFE_WR

SYS_Vss1v2kS_AFEAGND

EN

SEL_ACQUIRE

VDD

VSS_NEG

GND

RHD

S1_CLAMP

S2_PROTECTED

SYS_Ed1Zw3Jh_AFE_INJ0_P

SYS_Vss1v2kS_AFEAGND

recording-front-end_RECOB_PROT

TMUX8219DGKR

U30109

SYS_EGeeYbbK_AFE_RX3

SYS_hmS69999_AFE_ACQUIRE

SYS_9tpdO5bk_AFE_PS

SYS_d7bCd2ew_AFE_WR

SYS_Vss1v2kS_AFEAGND

EN

SEL_ACQUIRE

VDD

VSS_NEG

GND

RHD

S1_CLAMP

S2_PROTECTED

SYS_aw75Ga8R_AFE_INJ0_P

SYS_Vss1v2kS_AFEAGND

recording-front-end_RECOA_PROT

TMUX8219DGKR

U30110

SYS_EGeeYbbK_AFE_RX3

SYS_hmS69999_AFE_ACQUIRE

SYS_9tpdO5bk_AFE_PS

SYS_d7bCd2ew_AFE_WR

SYS_Vss1v2kS_AFEAGND

EN

SEL_ACQUIRE

VDD

VSS_NEG

GND

RHD

S1_CLAMP

S2_PROTECTED

SYS_BKaYcVA4_AFE_INJ1_P

SYS_Vss1v2kS_AFEAGND

recording-front-end_RECS3_PROT

TMUX8219DGKR

U30111

SYS_EGeeYbbK_AFE_RX3

SYS_hmS69999_AFE_ACQUIRE

SYS_9tpdO5bk_AFE_PS

SYS_d7bCd2ew_AFE_WR

SYS_Vss1v2kS_AFEAGND

EN

SEL_ACQUIRE

VDD

VSS_NEG

GND

RHD

S1_CLAMP

S2_PROTECTED

SYS_bKaYcVB7_AFE_INJ1_P

SYS_Vss1v2kS_AFEAGND

recording-front-end_RECS2_PROT

TMUX8219DGKR

C30108

SYS_Ed1Zw3Jh_AFE_INJ0_P

470p C0G 50V

SYS_Vss1v2kS_AFEAGND

C30109

SYS_aw75Ga8R_AFE_INJ0_P

470p C0G 50V

SYS_Vss1v2kS_AFEAGND

C30110

SYS_BKaYcVA4_AFE_INJ1_P

470p C0G 50V

SYS_Vss1v2kS_AFEAGND

C30111

SYS_bKaYcVB7_AFE_INJ1_P

470p C0G 50V

SYS_Vss1v2kS_AFEAGND

C30208

SYS_9tpdO5bk_AFE_PS

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30209

SYS_9tpdO5bk_AFE_PS

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30210

SYS_9tpdO5bk_AFE_PS

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30211

SYS_9tpdO5bk_AFE_PS

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30308

SYS_d7bCd2ew_AFE_WR

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30309

SYS_d7bCd2ew_AFE_WR

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30310

SYS_d7bCd2ew_AFE_WR

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

C30311

SYS_d7bCd2ew_AFE_WR

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2; S1 grounds RHD until ACQUIRE is admitted.

C30022

SYS_9tpdO5bk_AFE_PS

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

U30022

SYS_3bcteeh7_AFE_CONNECT1

SYS_3bcteeh7_AFE_CONNECT2

SYS_3bcteeh7_AFE_CONNECT3

SYS_3bcteeh7_AFE_CONNECT4

SYS_9tpdO5bk_AFE_PS

SYS_d7bCd2ew_AFE_WR

SYS_Vss1v2kS_AFEAGND

EP_SHIELD

SEL1

SEL2

SEL3

SEL4

VDD

VSS_NEG

GND

S1_CONTACT

S1_INTERNAL

S2_CONTACT

S2_INTERNAL

S3_CONTACT

S3_INTERNAL

S4_CONTACT

S4_INTERNAL

FF_N

recording-front-end_RECOB_SERVE1

recording-front-end_RECOB_PROT

recording-front-end_RECOA_SERVE1

recording-front-end_RECOA_PROT

recording-front-end_RECS3_SERVE1

recording-front-end_RECS3_PROT

recording-front-end_RECS2_SERVE1

recording-front-end_RECS2_PROT

recording-front-end_AFE_FAULT_N

TMUX7612FRRPR

C30032

SYS_d7bCd2ew_AFE_WR

100n X7R 16V

SYS_Vss1v2kS_AFEAGND

470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /recording-front-end/bank2/

File: bank2.kicad_sch

Title: ATO S4 / Inputs 08–11 / protected recording

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 15/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

R30112
recording-front-end_REC12_CONTACT
1
R
2
recording-front-end_REC12_SERIES
4.99k 0.1%

R30113
recording-front-end_REC13_CONTACT
1
R
2
recording-front-end_REC13_SERIES
4.99k 0.1%

R30114
recording-front-end_REC14_CONTACT
1
R
2
recording-front-end_REC14_SERIES
4.99k 0.1%

R30115
recording-front-end_REC15_CONTACT
1
R
2
recording-front-end_REC15_SERIES
4.99k 0.1%

R30212
recording-front-end_REC12_PROT
1
R
2
SYS_hia12k5_AFEAGND
10M

R30213
recording-front-end_REC13_PROT
1
R
2
SYS_hia12k5_AFEAGND
10M

R30214
recording-front-end_REC14_PROT
1
R
2
SYS_hia12k5_AFEAGND
10M

R30215
recording-front-end_REC15_PROT
1
R
2
SYS_hia12k5_AFEAGND
10M

U30112
SYS_65aef666_AFE_RX5
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da05066_AFE_PS
SYS_d7bc2dee_AFE_AR5
SYS_hia12k5_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
RHD
S1_CLAMP
S2_PROTECTED
1
2
8
SYS_hhbccccb_AFE_RX12_P
recording-front-end_REC12_PROT
TMUX6219DGKR

U30113
SYS_65aef666_AFE_RX5
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da05066_AFE_PS
SYS_d7bc2dee_AFE_AR5
SYS_hia12k5_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
RHD
S1_CLAMP
S2_PROTECTED
1
2
8
SYS_52c7e11c_AFE_RX13_P
SYS_hia12k5_AFEAGND
recording-front-end_REC13_PROT
TMUX6219DGKR

U30114
SYS_65aef666_AFE_RX5
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da05066_AFE_PS
SYS_d7bc2dee_AFE_AR5
SYS_hia12k5_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
RHD
S1_CLAMP
S2_PROTECTED
1
2
8
SYS_4f585988_AFE_RX14_P
SYS_hia12k5_AFEAGND
recording-front-end_REC14_PROT
TMUX6219DGKR

U30115
SYS_65aef666_AFE_RX5
SYS_4e6b5999_AFE_ACQUIRE
SYS_9da05066_AFE_PS
SYS_d7bc2dee_AFE_AR5
SYS_hia12k5_AFEAGND
EN
SEL_ACQUIRE
VDD
VSS_NEG
GND
RHD
S1_CLAMP
S2_PROTECTED
1
2
8
SYS_53213f89_AFE_RX15_P
SYS_hia12k5_AFEAGND
recording-front-end_REC15_PROT
TMUX6219DGKR

C30112
SYS_hhbccccb_AFE_RX12_P
1
C
2
470p C0G 50V

C30113
SYS_52c7e11c_AFE_RX13_P
1
C
2
SYS_hia12k5_AFEAGND
470p C0G 50V

C30114
SYS_4f585988_AFE_RX14_P
1
C
2
SYS_hia12k5_AFEAGND
470p C0G 50V

C30115
SYS_53213f89_AFE_RX15_P
1
C
2
SYS_hia12k5_AFEAGND
470p C0G 50V

C30212
SYS_9da05066_AFE_PS
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30213
SYS_9da05066_AFE_PS
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30214
SYS_9da05066_AFE_PS
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30215
SYS_9da05066_AFE_PS
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30312
SYS_d7bc2dee_AFE_AR5
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30313
SYS_d7bc2dee_AFE_AR5
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30314
SYS_d7bc2dee_AFE_AR5
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

C30315
SYS_d7bc2dee_AFE_AR5
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2; S1 grounds RHD until ACQUIRE is admitted.

C30023
SYS_9da05066_AFE_PS
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

U30023
SYS_3bcbeeb7_AFE_CONNECT
SYS_3bcbeeb7_AFE_CONNECT
SYS_3bcbeeb7_AFE_CONNECT
SYS_3bcbeeb7_AFE_CONNECT
SYS_9da05066_AFE_PS
SYS_d7bc2dee_AFE_AR5
SYS_hia12k5_AFEAGND
SYS_hia12k5_AFEAGND
SEL1
SEL2
SEL3
SEL4
VDD
VSS_NEG
GND
EP_SHIELD
S1_CONTACT
S1_INTERNAL
S2_CONTACT
S2_INTERNAL
S3_CONTACT
S3_INTERNAL
S4_CONTACT
S4_INTERNAL
FF_R
recording-front-end_REC12_SERIES
recording-front-end_REC12_PROT
recording-front-end_REC13_SERIES
recording-front-end_REC13_PROT
recording-front-end_REC14_SERIES
recording-front-end_REC14_PROT
recording-front-end_REC15_SERIES
recording-front-end_REC15_PROT
recording-front-end_AFE_FAULT_N
TMUX7412FRRPR

C30033
SYS_d7bc2dee_AFE_AR5
1
C
2
SYS_hia12k5_AFEAGND
100n X7R 16V

470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

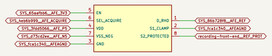
Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain
Project ATO
Sheet: /recording-front-end/bank3/
File: bank3.kicad_sch
Title: ATO S4 / Inputs 12–15 / protected recording
Size: A1 | Date: 2026-10-13 | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 16/80

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

REF_CONTACT passes through the same disconnect/clamp topology; one AFE_REF node feeds all sixteen minus pins (192pF intrinsic typical).











BIAS_CONTACT is separate and never a stimulation return. Passive current arithmetic does not establish a medical current limit.













Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /recording-front-end/reference/

File: reference.kicad_sch

Title: ATO S4 / Shared reference / passive body bias

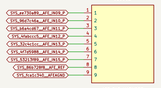
Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 17/80

INTEGRATED COMPONENT–TAP HARNESS / recording-front-end

Physical headers expose actual earlier component taps. Original connectors remain available.

Only named electrical nets connect. AFE, HOST and companion grounds remain distinct.



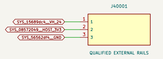


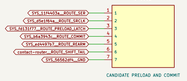
Artificial-load digital design. Added wiring capacitance and connector integrity require measurement.

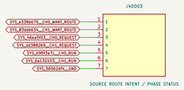
Project ATO
Sheet: /recording-front-end/integrated_component_taps/
File: integration_taps.kicad_sch
Title: ATO / recording-front-end integration taps
Size: A1 | Date: 2026-10-14 | Rev: 0.1
KiCad E.D.A. 10.0.6 | Id: 18/80

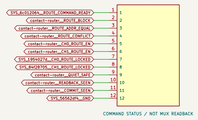
DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

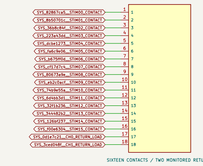












Required before COMMIT: both WANT/REQUEST/RUN low, independently observed quiescence, monitor permits, rails, recording isolation and watchdog valid.

Then: COMMIT -> read all eight active Q bits -> fresh verification ACK edge -> separate REARM -> assert route WANT -> allow settling before phase.

source0

File: source0.kicad_sch

source1

File: source1.kicad_sch

address

File: address.kicad_sch

admission

File: admission.kicad_sch

collision

File: collision.kicad_sch

No mux state sensor. Independent terminal cut-off, compliance detector, independent watchdog, rail generator. PCB or implanted device is completed here.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /contact-router/

File: contact_router.kicad_sch

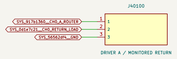
Title: ATO S4 / Route interfaces and monitored returns

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 19/80

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

Both bank 0 commons share A_ROUTER_END = ROUTE_EN & MODE0; EN1 = ROUTE_EN & ADDR3. Addresses 000..111 select S1..S8.



















Dedicated return: external contact -> RETURN_LOAD -> existing monitor 2 ohm shunt -> RETURN_DRIVER -> current-driver B_BR.

Do not bypass the monitor, add a second shunt, tie the two returns, or use recording REF/BIAS as a stimulation return.

TMUX8208 is rail-to-rail at valid 24 V supply: powered-off, stuck-on, parasitic coupling and two-source shared tissue paths remain unqualified.

A_ROUTER needs external terminal guard qualification before connection to A_BR; internal enable-low is only a commanded off state.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /contact-router/source0/

File: source0.kicad_sch

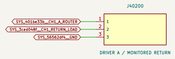
Title: ATO S4 / Source 0 / sixteen-contact output

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 20/80

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

Both bank 0 commons share A_ROUTER_END = ROUTE_EN & MODE0; EN1 = ROUTE_EN & ADDR3. Addresses 000..111 select S1..S8.

























Dedicated return: external contact -> RETURN_LOAD -> existing monitor 2 ohm shunt -> RETURN_DRIVER -> current-driver B_BR.

Do not bypass the monitor, add a second shunt, tie the two returns, or use recording REF/BIAS as a stimulation return.

TMUX8208 is rail-to-rail at valid 24 V supply: powered-off, stuck-on, parasitic coupling and two-source shared tissue paths remain unqualified.

A_ROUTER needs external terminal guard qualification before connection to A_BR; internal enable-low is only a commanded off state.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /contact-router/source1/

File: source1.kicad_sch

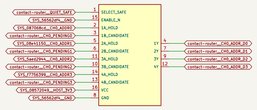
Title: ATO S4 / Source 1 / sixteen-contact output

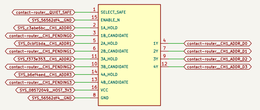
Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 21/80

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.













Candidate bits may change without changing active addresses. Eight active flip-flops always see raw COMMIT clock; safe D selection loads or holds.

RESET_READY low asynchronously clears active bits to zero; outputs remain inhibited until legal COMMIT, fresh readback ACK and deliberate REARM.

























SRCLR affects only the candidate shift stage; QH_SHIFT is payload cascade, not active-address or analog switch readback. Actual Q bits appear at J6.

QUIET_SAFE and candidate data must meet physical setup/hold. Gate delay, asynchronous clear, metastability and clock-integrity qualification remain.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /contact-router/address/

File: address.kicad_sch

Title: ATO S4 / Candidate preload and active addresses

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 22/80

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

BASE_HEALTH = RESET & RAILS & WATCHDOG & AFE_ISOLATED_READY & MONO & MON1; QUIET_SAFE additionally requires both WANT/REQUEST/RUN low and quiescence.

U40300

SYS_PFthz58__RESET_READY

SYS_B37U7R7__RAILS_QUALIFIED

SYS_wKtnGr4u__WATCHDOG_OK

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G110DVR

contact-router__HEALTH_A

U40301

SYS_SB52J17k__AFE_ISOLATED_READY

SYS_5bnbz13s__CH1_MON_PERMIT

SYS_xx31Lw8L__CH1_MON_PERMIT

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G110DVR

contact-router__HEALTH_B

U40302

contact-router__HEALTH_A

contact-router__HEALTH_B

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G08DBVR

contact-router__BASE_HEALTH

U40303

SYS_aT3Rb67S__CH1_WANT_ROUTE

SYS_B3ab6k7s__CH1_WANT_ROUTE

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G32DBVR

contact-router__ANY_WANT

U40304

contact-router__ANY_WANT

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G04DBVR

contact-router__NO_WANT

U40305

SYS_aT3Rb67S__CH1_REQUEST

SYS_eb3RE2Rk__CH1_REQUEST

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G32DBVR

contact-router__ANY_REQUEST

U40306

SYS_k99P3e7r__CH1_RUN

SYS_Kte15I18__CH1_RUN

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G32DBVR

contact-router__ANY_RUN

U40307

contact-router__ANY_REQUEST

contact-router__ANY_RUN

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G32DBVR

contact-router__ANY_PHASE

U40308

contact-router__ANY_PHASE

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G04DBVR

contact-router__NO_PHASE

U40309

SYS_2o0eSbJx__CH1_QUIESCENT_OK

SYS_2Gee4hJ__CH1_QUIESCENT_OK

contact-router__BASE_HEALTH

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G110DVR

contact-router__QUIET_HEALTH

U40310

contact-router__NO_WANT

contact-router__NO_PHASE

contact-router__QUIET_HEALTH

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G11DBVR

contact-router__QUIET_SAFE

U40311

SYS_b6a39t7c__ROUTE_COMMIT

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G04DBVR

contact-router__COMMIT_A

U40312

contact-router__ROUTE_CONFLICT

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G04DBVR

contact-router__NO_CONFLICT

U40313

contact-router__BASE_HEALTH

contact-router__COMMIT_A

contact-router__NO_CONFLICT

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G110DVR

contact-router__VALID_CLK_N

U40314

SYS_a6a7E5Jx__ROUTE_COMMIT

contact-router__QUIET_SAFE

contact-router__BASE_HEALTH

SYS_SB57Z0t4__HOST_N25

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G74DCUR

contact-router__COMMIT_SEEN

U40315

contact-router__BASE_HEALTH

contact-router__COMMIT_A

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G08DBVR

contact-router__ACK_CLR_N

U40316

SYS_1hP7G2L__ROUTE_ACTIVE_BITS_VERIFIED

contact-router__COMMIT_SEEN

contact-router__ACK_CLR_N

SYS_SB57Z0t4__HOST_N25

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G74DCUR

contact-router__READBACK_SEEN

U40317

contact-router__COMMIT_SEEN

contact-router__QUIET_SAFE

contact-router__READBACK_SEEN

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G110DVR

contact-router__READBACK

U40318

SYS_ga4n07k7__ROUTE_READBK

contact-router__READBACK

contact-router__BASE_HEALTH

contact-router__VALID_CLK_N

SYS_SB57Z0t4__HOST_N25

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G74DCUR

contact-router__ROUTE_VALID

U40319

contact-router__ROUTE_VALID

contact-router__BASE_HEALTH

contact-router__COMMIT_A

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G110DVR

contact-router__READY_A

U40320

contact-router__READY_A

contact-router__NO_CONFLICT

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G08DBVR

SYS_BcCL2Q6s__ROUTE_COMMAND_READY

U40321

contact-router__ROUTE_VALID

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

SN74LVC1G04DBVR

contact-router__ROUTE_BLOCK

C40300

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

100n X7R 10V

C40301

SYS_5656Z4PL__GND

SYS_SB57Z0t4__HOST_N25

100n X7R 10V

C40302

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40303

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40304

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40305

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40306

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40307

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40308

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40309

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

100n X7R 10V

C40310

SYS_5656Z4PL__GND

SYS_SB57Z0t4__HOST_N25

100n X7R 10V

C40311

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40312

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40313

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40314

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40315

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40316

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40317

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40318

SYS_SB57Z0t4__HOST_N25

SYS_5656Z4PL__GND

100n X7R 10V

C40319

SYS_5656Z4PL__GND

SYS_SB57Z0t4__HOST_N25

100n X7R 10V

C40320

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

C40321

SYS_5656Z4PL__GND

SYS_5656Z4PL__GND

100n X7R 10V

R40300

SYS_PFthz58__RESET_READY

10k

R40301

SYS_B37U7R7__RAILS_QUALIFIED

10k

R40302

SYS_wKtnGr4u__WATCHDOG_OK

10k

R40303

SYS_SB52J17k__AFE_ISOLATED_READY

10k

R40304

SYS_2o0eSbJx__CH1_QUIESCENT_OK

10k

R40305

SYS_1hP7G2L__ROUTE_ACTIVE_BITS_VERIFIED

10k

R40306

SYS_5bnbz13s__CH1_MON_PERMIT

10k

R40307

SYS_xx31Lw8L__CH1_MON_PERMIT

10k

R40308

SYS_5656Z4PL__GND

10k

R40309

SYS_aT3Rb67S__CH1_WANT_ROUTE

10k

R40310

SYS_aT3Rb67S__CH1_REQUEST

10k

R40311

SYS_aT3Rb67S__CH1_RUN

10k

R40312

SYS_k99P3e7r__CH1_RUN

10k

R40313

SYS_5656Z4PL__GND

10k

R40314

SYS_5656Z4PL__GND

10k

Any COMMIT invalidates prior validation and ACK. Illegal COMMIT holds active bits and captures COMMIT_SEEN=0.

Rearm needs a fresh post-commit ACK edge. Health recovery or collision removal cannot create a clock edge or restart output.

Ideal logic only: setup/hold, all-off settling, voltage, independent watchdog and actual current observation remain mandatory.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /contact-router/admission/

File: admission.kicad_sch

Title: ATO S4 / Commit, acknowledgement and rearm

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 23/80

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

Conflict = WANT0 & WANT1 & equal(all four active address bits). It disables BOTH bank families and asynchronously clears route validation.

U40400

SYS_5F7568cd_~{CH1_ADDR0}
SYS_c7abe0cc_~{CH1_ADDR1}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G86DBVR
contact-router_~{ADDR_DIFF0}

U40401

SYS_08e41150_~{CH0_ADDR0}
SYS_0c5f1b2a_~{CH0_ADDR1}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G86DBVR
contact-router_~{ADDR_DIFF1}

U40402

SYS_5ae2f914_~{CH0_ADDR2}
SYS_f375e703_~{CH1_ADDR2}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G86DBVR
contact-router_~{ADDR_DIFF2}

U40403

SYS_f7756398_~{CH0_ADDR3}
SYS_b6efaeed_~{CH1_ADDR3}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G86DBVR
contact-router_~{ADDR_DIFF3}

U40404

contact-router_~{ADDR_DIFF3}
contact-router_~{ADDR_DIFF1}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G32DBVR
contact-router_~{ADDR_DIFF_10}

U40405

contact-router_~{ADDR_DIFF2}
contact-router_~{ADDR_DIFF0}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G32DBVR
contact-router_~{ADDR_DIFF_01}

U40406

contact-router_~{ADDR_DIFF_10}
contact-router_~{ADDR_DIFF_01}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G32DBVR
contact-router_~{ADDR_ANY_DIFF}

U40407

contact-router_~{ADDR_ANY_DIFF}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G04DBVR
contact-router_~{ROUTE_ADDR_EQUAL}

U40408

SYS_a3586a75_~{CH0_WANT_ROUTE}
SYS_83ab6d3c_~{CH1_WANT_ROUTE}
contact-router_~{ROUTE_ADDR_EQUAL}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G11DBVR
contact-router_~{ROUTE_CONFLICT}

U40409

SYS_6c512264_~{ROUTE_COMMAND_READY}
SYS_a3586a75_~{CH0_WANT_ROUTE}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
contact-router_~{CH0_ROUTE_EN}

U40410

SYS_f7756398_~{CH0_ADDR3}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G04DBVR
contact-router_~{CH0_BANK3_SELECT}

U40411

contact-router_~{CH0_ROUTE_EN}
contact-router_~{CH0_BANK3_SELECT}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
contact-router_~{CH0_BANK3_EN}

U40412

contact-router_~{CH0_ROUTE_EN}
SYS_f7756398_~{CH0_ADDR3}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
contact-router_~{CH0_BANK3_EN}

U40413

SYS_6c512264_~{ROUTE_COMMAND_READY}
SYS_a3586a75_~{CH0_WANT_ROUTE}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
SYS_19543276_~{CH0_ROUTE_LOCKED}

U40414

SYS_6c512264_~{ROUTE_COMMAND_READY}
SYS_83ab6d3c_~{CH1_WANT_ROUTE}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
contact-router_~{CH1_ROUTE_EN}

U40415

SYS_b6efaeed_~{CH1_ADDR3}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G04DBVR
contact-router_~{CH1_BANK3_SELECT}

U40416

contact-router_~{CH1_ROUTE_EN}
contact-router_~{CH1_BANK3_SELECT}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
contact-router_~{CH1_BANK3_EN}

U40417

contact-router_~{CH1_ROUTE_EN}
SYS_b6efaeed_~{CH1_ADDR3}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
contact-router_~{CH1_BANK3_EN}

U40418

SYS_6c512264_~{ROUTE_COMMAND_READY}
SYS_83ab6d3c_~{CH1_WANT_ROUTE}
SYS_08572048_~{HOST_IN3}
SYS_56562df4_GND
SN74LVC1G08DBVR
SYS_84f28705_~{CH1_ROUTE_LOCKED}

C40400

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40401

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40402

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40403

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40404

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40405

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40406

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40407

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40408

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40409

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40410

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40411

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40412

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40413

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40414

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40415

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40416

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40417

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

C40418

SYS_08572048_~{HOST_IN3}
1
2
SYS_56562df4_GND
100n X7R 10V

Removing a conflict only removes its fault condition: it does not restore VALID. Release both WANTs, verify quiet, then explicitly rearm.

The two source enables may coexist only at different contacts. Common tissue impedance and volume conduction are outside this digital collision test.

ROUTE_LOCKED means admitted digital command ownership only. No selected-contact continuity or power-off isolation is inferred.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /contact-router/collision/

File: collision.kicad_sch

Title: ATO S4 / Conflict and bank enables

Size: A1	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 24/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.

J50001

- SYS_a4932174__SIGP_3V3
- SYS_68eea99b__SIGP_A
- SYS_88872d4b__HOST_3V3
- SYS_ce1d4858__HOST_A
- SYS_fb223756__ON_3V3
- SYS_1f68f6b3__ON_3V3
- SYS_56562d9c__GND

EXTERNAL SUPERVISOR AND HOST RAILS

J50002

- SYS_f3ee9bbb__AFE_3V3
- SYS_39d5056d__AFE_5V
- SYS_af7b1d4e__AFE_A5
- SYS_7ce17e6_AFE_ADJ_GND
- SYS_8bc1ee7c__AFE_CONNECT
- SYS_ea66b998__AFE_ACQUIRE

FLOATING AFE RAILS AND OBSERVED CONTROLS

J50003

- SYS_65555e6e__SIGP_REARM
- SYS_82553898__FAULT_RESET
- SYS_29baee1c_WD
- SYS_48e45f03__CHA_REQUEST
- SYS_ed388305__CHJ_REQUEST
- SYS_69f0267c__CHA_ACK
- SYS_6e132101__CHJ_ACK
- SYS_1f47477a__CHA_ROUTE_LOCKED
- SYS_8429f705__CHJ_ROUTE_LOCKED
- SYS_7d38bc21__PHASE_REARM
- SYS_48e29157__PHASE_REARM
- SYS_5d562d9c__GND

RAW CONTROL / STATUS

J50004

- SYS_9be3b73__CHA_MON
- SYS_51b24a4c__CHJ_MON
- SYS_57b982a__MON_VBUS
- SYS_48827f2__MON_VBAT
- SYS_58ebc371__CHA_MON_PGOOD
- SYS_ce7312eb__CHJ_MON_PGOOD
- SYS_43ee119a__CHA_EXT_SUP_OK
- SYS_bafa894a__CHJ_EXT_SUP_OK
- SYS_5d562d9c__GND

EXISTING MONITOR OBSERVATIONS

J50005

- SYS_4f7bc58__AFE_ET_READY
- SYS_93b17b57__BUS_SIMUFIED
- SYS_a83d49fc__WATCHDOG_OK
- SYS_8d5217e__AFE_DISCARTE_READY
- SYS_2b9e5b2e__CHA_QUIESCENT_OK
- SYS_20bbe6c1__CHJ_QUIESCENT_OK
- SYS_e3b7c7fc__CHA_EXT_SUP_OK
- SYS_8f0f6fce__CHJ_EXT_SUP_OK
- SYS_98c50981__SUP_ARMED
- supervisor__CHA_ACK
- supervisor__CHJ_ACK
- SYS_7fc27e0d__GLOBAL_GUARD_OK
- SYS_5d562d9c__GND

EXPORTED NATIVE HEALTH / PERMISSION

J50006

- SYS_8175178b__CHA_A_ROUTE
- SYS_ee17d917__CHA_B_ROUTE
- SYS_c651ee7b__CHJ_A_ROUTE
- SYS_c7314b6c__CHJ_B_RET
- SYS_5d562d9c__GND

DRIVER INTERNAL DIFFERENTIAL VOLTAGES

SYS_a4932174__SIGP_3V3

SYS_68eea99b__SIGP_A

guards

File: guards.kicad_sch

afe_rails

File: afe_rails.kicad_sch

quiescence

File: quiescence.kicad_sch

admission

File: admission.kicad_sch

phase1

File: phase1.kicad_sch

afe_bank2

File: afe_bank2.kicad_sch

afe_logic

File: afe_logic.kicad_sch

host_rails

File: host_rails.kicad_sch

afe_rails_2

File: afe_rails_2.kicad_sch

voltage

File: voltage.kicad_sch

admission_2

File: admission_2.kicad_sch

afe_bank0

File: afe_bank0.kicad_sch

afe_bank3

File: afe_bank3.kicad_sch

afe_logic_2

File: afe_logic_2.kicad_sch

host_rails_2

File: host_rails_2.kicad_sch

watchdog

File: watchdog.kicad_sch

duration

File: duration.kicad_sch

phase0

File: phase0.kicad_sch

afe_bank1

File: afe_bank1.kicad_sch

afe_bank4

File: afe_bank4.kicad_sch

defaults

File: defaults.kicad_sch

integrated_component_taps

File: integration_taps.kicad_sch

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/

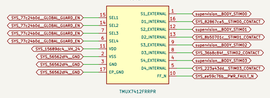
File: supervision.kicad_sch

Title: ATO S4 / interfaces and handoff

Size: A1	Date: 2026-10-14
KiCad E.D.A. 10.0.6	**Rev: 0.1**
	Id: 25/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.















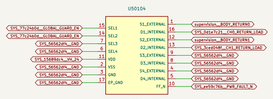






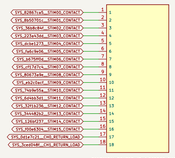




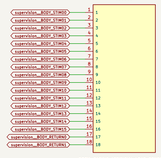








CANONICAL ROUTER J7 // BEFORE GUARDS



BODY-FACING ARTIFICIAL LOAD CONTACTS



All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/guards/

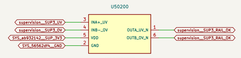
File: guards.kicad_sch

Title: ATO S4 / 18 body-side terminal guards

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 26/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.





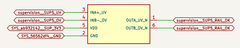






































All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/host_rails/

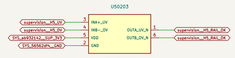
File: host_rails.kicad_sch

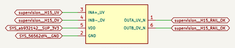
Title: ATO S4 / Host supply windows

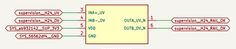
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 27/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.









All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/host_rails_2/

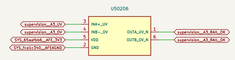
File: host_rails_2.kicad_sch

Title: ATO S4 / Host supply windows / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 28/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.





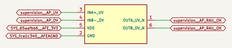














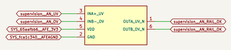














All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_rails/

File: afe_rails.kicad_sch

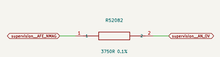
Title: ATO S4 / AFE supply windows

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 29/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.































All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_rails_2/

File: afe_rails_2.kicad_sch

Title: ATO S4 / AFE supply windows / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 30/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.

R50250
100k 1%
SYS_ads9327x2__SIGP_SCK
supervision__SIGP_SCK
C50250
47n X7R 10%
supervision__SIGP_SCK
SYS_5050249%__GND
D50250
BAT54WS-7-F
supervision__SIGP_SCK
supervision__SIGP_SDI
R50251
100R
supervision__SIGP_SDI
SYS_ads9327x2__SIGP_SDI
U50250
SN74LVC1G17DBVR
supervision__SIGP_SCK
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__SIGP_STARTUP_CH
C50251
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51000
SN74LVC1G110DBVR
supervision__SIGP3_RAIL_OK
supervision__SIGP5_RAIL_OK
supervision__5V3_RAIL_OK
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__HOST_RAIL5_OK_LS
C51000
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51001
SN74LVC1G110DBVR
supervision__HOST_RAIL5_OK_LT
supervision__HV_RAIL_OK
supervision__5V3_RAIL_OK
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__HOST_RAIL5_OK_LS
C51001
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51002
SN74LVC1G110DBVR
supervision__HOST_RAIL5_OK_LT
supervision__HOST_RAIL5_OK
supervision__SIGP_STARTUP_CH
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__HOST_RAIL5_OK
C51002
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U50300
TPS2531DRBR
SYS_ads9327x2__SIGP_SDI
supervision__WDI_CW2
supervision__HOST_RAIL5_OK
SYS_5050249%__GND
SYS_aTPower3__WDI
SYS_5050249%__GND
WDO_FAILING
CP_GND
WDO_B
ENOUT
supervision__WDI_ENOUT_CH
supervision__WDI_ENOUT_CH
R50300
10k
SYS_ads9327x2__SIGP_SDI
supervision__WDI_CW2
R50301
10k
SYS_ads9327x2__SIGP_SDI
supervision__WDI_ENOUT_CH
R50302
10k
SYS_aTPower3__WDI
SYS_5050249%__GND
C50300
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51005
SN74LVC1G04DBVR
SYS_aTPower3__WDI
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__WDI_B
C51005
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51006
SN74LVC1G08DBVR
supervision__WDI_ENOUT_CH
supervision__HOST_RAIL5_OK
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__WDI_SEEN_CLR_N
U50301
SN74LVC1G74DCUR
supervision__WDI_B
supervision__WDI_ENOUT_CH
supervision__WDI_SEEN_CLR_N
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__WDI_SEEN
C50301
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51007
SN74LVC1G11DBVR
supervision__WDI_ENOUT_CH
supervision__WDI_SEEN
supervision__HOST_RAIL5_OK
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
supervision__WDI_GOOD
C51007
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
U51008
SN74LVC1G08DBVR
supervision__WDI_GOOD
SYS_ads9327x2__SIGP_SDI
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND
SYS_aB5049x__WATCHDOG_OK
C51008
100n X7R 16V
SYS_ads9327x2__SIGP_SDI
SYS_5050249%__GND

All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain
Project ATO
Sheet: /supervision/watchdog/
File: watchdog.kicad_sch
Title: ATO S4 / Heartbeat and startup
Size: A1 | Date: 2026-10-14 | Rev: 0.1
KiCad E.D.A. 10.0.6 | Id: 31/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.









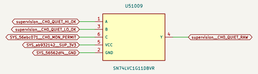












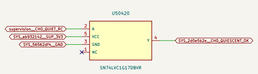




















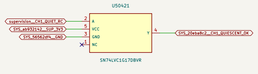














All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/quiescence/

File: quiescence.kicad_sch

Title: ATO S4 / Current quiet window

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 32/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.





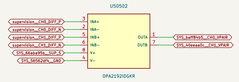












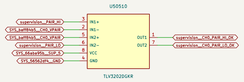




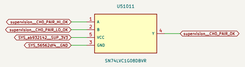


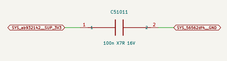










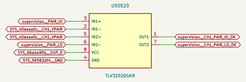




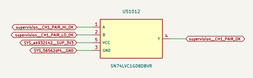














All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/voltage/

File: voltage.kicad_sch

Title: ATO S4 / Differential compliance

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 33/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.























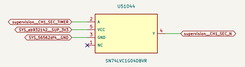








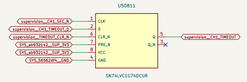














All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/duration/

File: duration.kicad_sch

Title: ATO S4 / Secondary phase duration

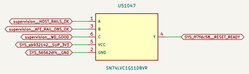
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 34/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.























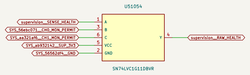








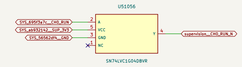












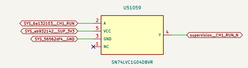












All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/admission/

File: admission.kicad_sch

Title: ATO S4 / Sticky permission and rearm

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 35/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.

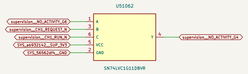




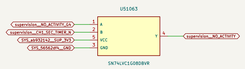




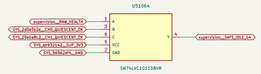






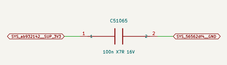












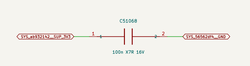




























All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/admission_2/

File: admission_2.kicad_sch

Title: ATO S4 / Sticky permission and rearm / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 36/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

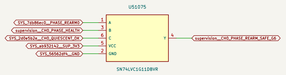
18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.

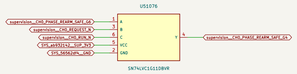




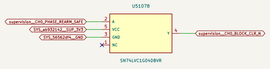


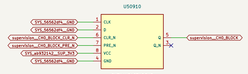


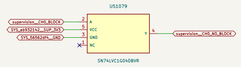






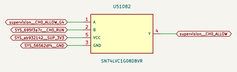


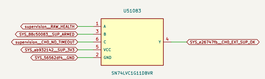












All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/phase0/

File: phase0.kicad_sch

Title: ATO S4 / Source 0 phase permission

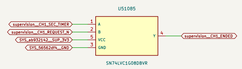
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 37/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.









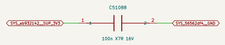


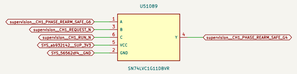




































All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain
Project ATO
Sheet: /supervision/phase1/
File: phase1.kicad_sch

Title: ATO S4 / Source 1 phase permission

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 38/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



















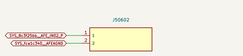






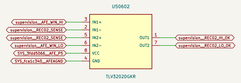




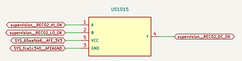


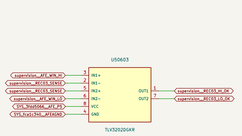


















All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_bank0/

File: afe_bank0.kicad_sch

Title: ATO S4 / RHD input windows 0

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 39/80

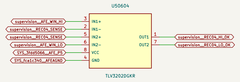
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

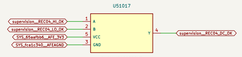




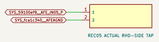






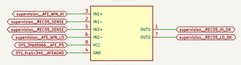








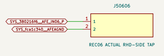






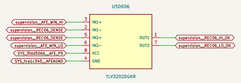






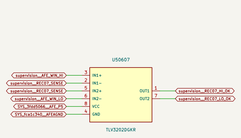


























All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_bank1/

File: afe_bank1.kicad_sch

Title: ATO S4 / RHD input windows 1

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 40/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.





































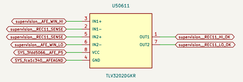


















All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_bank2/

File: afe_bank2.kicad_sch

Title: ATO S4 / RHD input windows 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 41/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

J50612

REC12 ACTUAL RHD-SIDE TAP

R56120

10M 0.1%

R56121

1M 0.1%

U50612

TLV320204GKR

C50612

100n X7R 16V

U51025

SN74LVC1G08DBVR

C51025

100n X7R 16V

J50613

REC13 ACTUAL RHD-SIDE TAP

R56130

10M 0.1%

R56131

1M 0.1%

U50613

TLV320204GKR

C50613

100n X7R 16V

U51026

SN74LVC1G08DBVR

C51026

100n X7R 16V

J50614

REC14 ACTUAL RHD-SIDE TAP

R56140

10M 0.1%

R56141

1M 0.1%

U50614

TLV320204GKR

C50614

100n X7R 16V

U51027

SN74LVC1G08DBVR

C51027

100n X7R 16V

J50615

REC15 ACTUAL RHD-SIDE TAP

R56150

10M 0.1%

R56151

1M 0.1%

U50615

TLV320204GKR

C50615

100n X7R 16V

U51028

SN74LVC1G08DBVR

C51028

100n X7R 16V

All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_bank3/

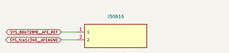
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Title: ATO S4 / RHD input windows 3

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 42/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

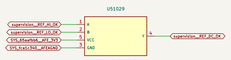














All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_bank4/

File: afe_bank4.kicad_sch

Title: ATO S4 / RHD input windows 4

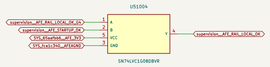
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 43/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.







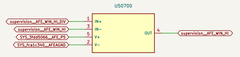




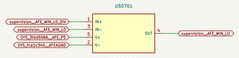




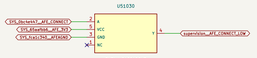




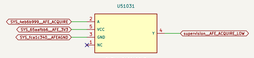






















All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_logic/

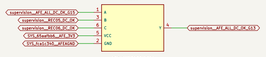
File: afe_logic.kicad_sch

Title: ATO S4 / AFE blank and isolated status

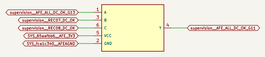
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 44/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

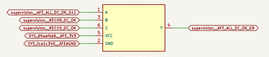
18 guarded stimulation conductors. 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.



















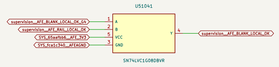




















All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/afe_logic_2/

File: afe_logic_2.kicad_sch

Title: ATO S4 / AFE blank and isolated status / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 45/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHO-side windows; conditional healthy-switch blanking is not proven isolation.

























All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision: converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /supervision/defaults/

File: defaults.kicad_sch

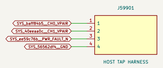
Title: ATO S4 / External defaults

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 46/80

INTEGRATED COMPONENT—TAP HARNESS / supervision

Physical headers expose actual earlier component taps. Original connectors remain available.

Only named electrical nets connect. AFE, HOST and companion grounds remain distinct.



Artificial-load digital design. Added wiring capacitance and connector integrity require measurement.

Project ATO
Sheet: /supervision/integrated_component_taps/
File: integration_taps.kicad_sch
Title: ATO / supervision integration taps
Size: A1 | Date: 2026-10-14 | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 47/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

J60001

power__PACK_POS
SYS_S6562GPL__GND

ATJ1004 PROTECTED 25 PACK / 6A TO 8.4V

J60002

power__PWR_REQUEST
power__SUP_READY
power__BATT_OK
power__HOST_ENABLE
power__BOOST_ENABLE
power__HOST_RAIL_PG
power__HOST_V_PG
power__SUP_RAIL_PG
power__SUP_V_PG
SYS_ak93215J__SUP_SCL
SYS_ak9567EG__PWR_FAULT_N
SYS_S6562GPL__GND

POWER REQUEST / STATUS

J60003

SYS_ak93215J__SUP_SCL
SYS_AS6562PGL__SUP_V
SYS_SM7120VA__HOST_SCL
SYS_ak4568CN__HOST_V
SYS_SJ22378__VA_23
SYS_LTC4089xx1__VA_24
SYS_S6562GPL__GND

HOST AND SUPERVISOR RAILS

J60004

SYS_AS6562GPL__PACK_POS
SYS_TPS7A2033__AFE_3V3
SYS_TPS7A2018__AFE_1V8
SYS_TPS7A2018__AFE_1V8
SYS_ak4376EG__AFE4GND

FLOATING RECORDING RAILS

J60006

SYS_AS6562GPL__MON_VBAT
SYS_TPS7A2033__MON_VMID
SYS_S6562GPL__GND

EXISTING MONITOR REFERENCES

J60005

SYS_PJ3968GPL__PACK_BUS
power__VBAT_P1
power__VCC_3V3
SYS_TLV1762__AFE4GND
SYS_S6562GPL__GND

CONVERTER TEST POINTS

power__PACK_POS

power__PACK_POS

SYS_PJ3968GPL__PACK_BUS

input

File: input.kicad_sch

host3

File: host3.kicad_sch

enable_2

File: enable_2.kicad_sch

boost24

File: boost24.kicad_sch

afe3

File: afe3.kicad_sch

sup3

File: sup3.kicad_sch

host5

File: host5.kicad_sch

enable_3

File: enable_3.kicad_sch

Island

File: Island.kicad_sch

afe4

File: afe4.kicad_sch

sup5

File: sup5.kicad_sch

enable

File: enable.kicad_sch

boost15

File: boost15.kicad_sch

afe3

File: afe3.kicad_sch

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/

File: power.kicad_sch

Title: ATO S4 / Interfaces and rail handoff

Size: A1	Date: 2026-10-14
KiCad E.D.A. 10.0.6	Rev: 0.1 Id: 48/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.











All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/input/

File: input.kicad_sch

Title: ATO S4 / Protected pack and input disconnect

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 49/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/sup3/

File: sup3.kicad_sch

Title: ATO S4 / Always-on supervisor 3.3 V

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 50/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/sup5/

File: sup5.kicad_sch

Title: ATO S4 / Always-on supervisor 5 V

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 51/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/host3/

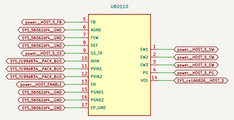
File: host3.kicad_sch

Title: ATO S4 / Host logic 3.3 V

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 52/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.





















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/host5/

File: host5.kicad_sch

Title: ATO S4 / Host and module input 5 V

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 53/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.





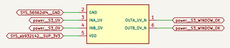




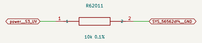










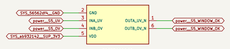






















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/enable/

File: enable.kicad_sch

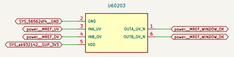
Title: ATO S4 / Native battery and startup qualification

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 54/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







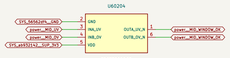


















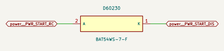
















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/enable_2/

File: enable_2.kicad_sch

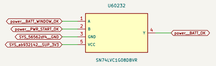
Title: ATO S4 / Native battery and startup qualification / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 55/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















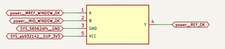


















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/enable_3/

File: enable_3.kicad_sch

Title: ATO S4 / Native battery and startup qualification / 3

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 56/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.























All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/boost15/
File: boost15.kicad_sch

Title: ATO S4 / Host 15 V boost

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 57/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.























All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/boost24/

File: boost24.kicad_sch

Title: ATO S4 / Stimulation 24 V boost

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 58/80

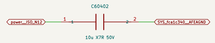
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.









All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

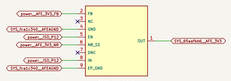
Sheet: /power/island/
File: island.kicad_sch

Title: ATO S4 / Floating recording power module

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 59/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







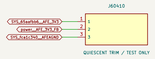












All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/afe3/

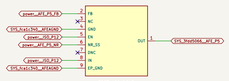
File: afe3.kicad_sch

Title: ATO S4 / Floating AFE 3.45 V regulator

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 60/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







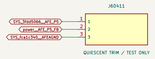












All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO

Sheet: /power/afep/

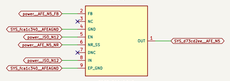
File: afep.kicad_sch

Title: ATO S4 / Floating AFE +5.30 V regulator

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 61/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







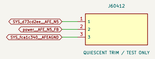












All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project ATO		
Sheet: /power/afen/		
File: afen.kicad_sch		
Title: ATO S4 / Floating AFE −5.10 V regulator		
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 62/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.

J70001

HOST RAILS / RAW ADC

J70002

EXISTING ANALOG TAPS

J70003

DEDICATED RAW ADC BUS

monitoring-transport_COMP_5V3

monitoring-transport_COMP_GND

root_2

File: root_2.kicad_sch

inputs

File: inputs.kicad_sch

handoff

File: handoff.kicad_sch

adc

File: adc.kicad_sch

health

File: health.kicad_sch

reference

File: reference.kicad_sch

companion

File: companion.kicad_sch

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/

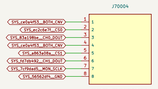
File: monitoring_transport.kicad_sch

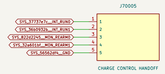
Title: ATO S4 / Existing package and host endpoints

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 63/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.







All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/root_2/

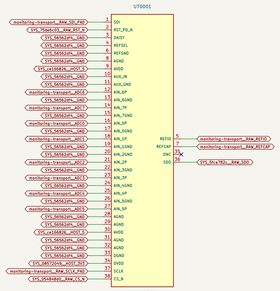
File: root_2.kicad_sch

Title: ATO S4 / Existing package and host endpoints / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 64/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.















All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/adc/

File: adc.kicad_sch

Title: ATO S4 / Dedicated eight-channel raw conversion

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 65/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.





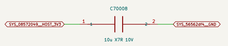














All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/reference/

File: reference.kicad_sch

Title: ATO S4 / independent ADC reference and bypass

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 66/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.

































All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/inputs/

File: inputs.kicad_sch

Title: ATO S4 / Raw current and scaled voltage filtering

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 67/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.









All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/health/

File: health.kicad_sch

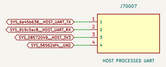
Title: ATO S4 / Divided power observations

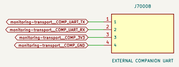
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 68/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.









All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/companion/

File: companion.kicad_sch

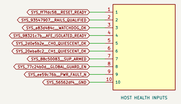
Title: ATO S4 / isolated wired processed-feature companion

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 69/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / MONITORING AND COMPANION

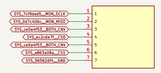
Dedicated raw current/voltage ADC and wired processed-feature companion. Endpoint charge ADCs remain separate.











All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native design: target firmware and PCB verification follow

Project ATO

Sheet: /monitoring-transport/handoff/

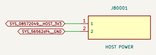
File: handoff.kicad_sch

Title: ATO S4 / Charge ADC and health host handoff

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 70/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.





processor

File: processor.kicad_sch

handoff

File: handoff.kicad_sch

defaults

File: defaults.kicad_sch

power

File: power.kicad_sch

handoff_2

File: handoff_2.kicad_sch

defaults_2

File: defaults_2.kicad_sch

sideband

File: sideband.kicad_sch

handoff_3

File: handoff_3.kicad_sch

defaults_3

File: defaults_3.kicad_sch

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller-carrier/

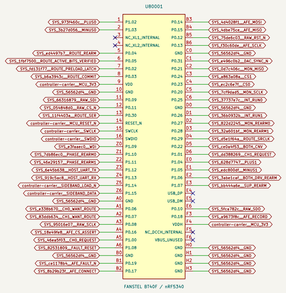
File: controller_carrier.kicad_sch

Title: ATO S4 / Controller carrier / interface index

Size: A1	Date: 2026-10-14
KiCad E.D.A. 10.0.6	**Rev: 0.1**
	Id: 71/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.



All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller--carrier/processor/

File: processor.kicad_sch

Title: ATO S4 / BT40F / actual nRF5340 physical pads

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 72/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.

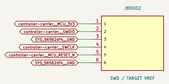














All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller-carrier/power/

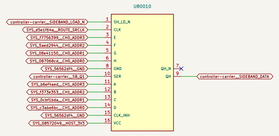
File: power.kicad_sch

Title: ATO S4 / Module normal-voltage supply and SWD

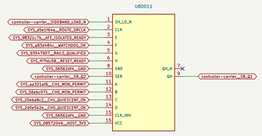
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 73/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

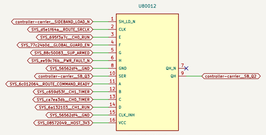
46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.



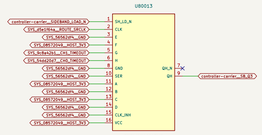














All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller-carrier/sideband/

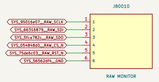
File: sideband.kicad_sch

Title: ATO S4 / Four physical 165 devices / parallel observations

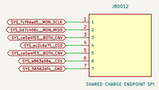
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 74/80

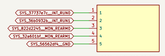
DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.









All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheets: /controller--carrier/handoff/

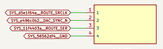
File: handoff.kicad_sch

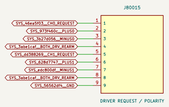
Title: ATO S4 / Matched peripheral handoff connectors

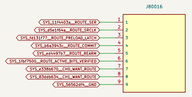
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 75/80

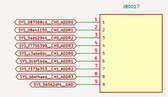
DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.









All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller--carrier/handoff_2/

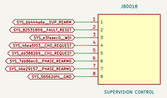
File: handoff_2.kicad_sch

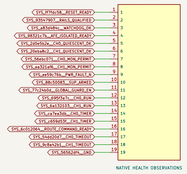
Title: ATO S4 / Matched peripheral handoff connectors / 2

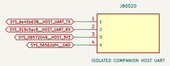
Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 76/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.







All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller-carrier/handoff_3/

File: handoff_3.kicad_sch

Title: ATO S4 / Matched peripheral handoff connectors / 3

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 77/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.





























All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheet: /controller-carrier/defaults/

File: defaults.kicad_sch

Title: ATO S4 / Passive reset defaults / explicit recovery

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 78/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.





























All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier: physical execution and assembly qualification remain

Project ATO

Sheet: /controller-carrier/defaults_2/

File: defaults_2.kicad_sch

Title: ATO S4 / Passive reset defaults / explicit recovery / 2

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 79/80

DIGITAL SCCI ENGINEERING / ARTIFICIAL SIGNALS / nRF5340 CONTROLLER CARRIER

46 actual GPIO, explicit common CNV/rearm and physical 32-bit health/readback chain. Radio remains disabled.

























All labels are native electrical connections. Digital artificial-load interface; no clinical, chronic or body-radio qualification.

Hardware permissions and terminal guards remain independent of transport. Loss of data rejects the observation.

Native module carrier; physical execution and assembly qualification remain

Project ATO

Sheets: /controller--carrier/defaults_3/

File: defaults_3.kicad_sch

Title: ATO S4 / Passive reset defaults / explicit recovery / 3

Size: A1	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 80/80